EFY TESTED
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COUNTING SMALL ITEMS
Using Edge Impulse

SOMNATH BERA

Object classification is essential for factory automation and devices where different objects are sorted or counted during packaging. Factories producing buttons, chocolates, and similar items require accurate counting for efficient packaging, which is a time-consuming and costly process when done manually. Implementing an automation system to detect, classify, sort, and count objects can significantly streamline production process. This project aims to create a cost-effective device for implementing such a factory automation system.

Object classifications are easily achievable using Edge Impulse models. You can distinguish between a man and animals, a bi-cycle and other types of vehicles, and so on. Similarly, you can accurately count one type of object among other types.

Initially designed for MCU level implementation on devices like ESP32 and Arduino Nicla Vision, the project was intended for counting a small area of 120 pixels x 120 pixels, with a relatively small-sized button as the object of interest. However, it became apparent that even for this small area, the MCUs were inadequate, as the model file itself is approximately 8MB long. Consequently, the project was eventually installed on a Raspberry Pi computer, where it operates seamlessly. Refer to Fig. 1 for an illustration of the author's working prototype.

The components required for the project are listed in Bill of Materials table.

Connection

Refer to Fig. 2 for camera connection to the Raspberry Pi. Connect the Raspberry Pi camera and display as shown in the picture and connect the HDMI display to the HDMI port of the Raspberry Pi.

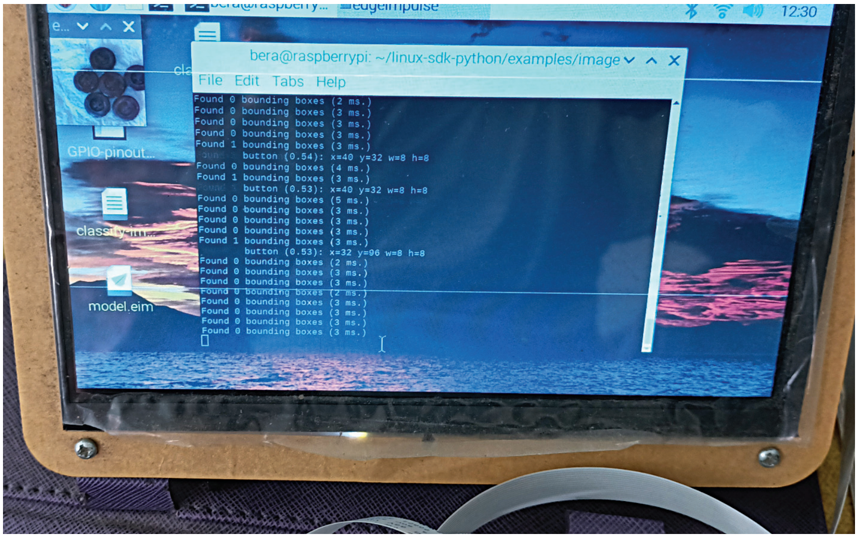

Fig. 1: Author's working prototype

BILL OF MATERIALS

Components	Quantity
Raspberry PI Zero W /4	1
RPi camera module	1
SD card 18Gb and above	1
HDMI display	1

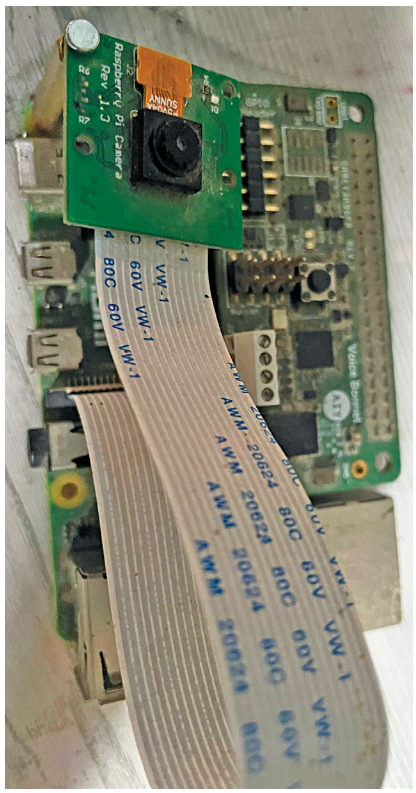

Fig. 2: Camera connection to Raspberry Pi

Creating ML model

To create the machine language (ML) model, you need to classify and recognise objects. Various platforms like TensorFlow, Edge Impulse, and Google Teachable can be used for this purpose.

Start by opening an account on edgeimpulse.com using an email ID. Collect a handful of similar types of buttons. If you access the site from a Raspberry Pi computer, use the camera to collect images of but-